

Yao Wang

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Profile

Environmental engineer studying sustainable waste management, landfill biogeochemistry, greenhouse-gas accounting and mitigation, and the long-term fate of plastics and microplastics. My work combines bottom-up inventories, interpretable machine learning, field observations, and process-based models to support lower-carbon and lower-pollution waste systems.

Professional Experience

- **Nanyang Technological University**, Postdoctoral Research Fellow, Feb 2026-present. Supervisor: Associate Professor Xunchang Fei.

Education

- **Nanyang Technological University**, Ph.D. in Environmental Engineering, Aug 2022-Jan 2026. Thesis: The biogeochemical dynamics of major elements in municipal solid waste landfills.
- **Nanyang Technological University**, M.Eng. in Environmental Engineering, Aug 2020-Jul 2022. Thesis: The degradation processes of municipal solid waste in landfills.
- **Tongji University**, B.S. in Environmental Science, Sep 2016-Jun 2020. Thesis: QSAR modeling of chlorinated paraffin in aquatic ecosystems.

Research Interests

- AI-driven decision-making for sustainable waste management.
- Greenhouse-gas accounting and MRV for waste systems.
- Climate-waste interactions and resilience of land-disposed waste.
- Fates of plastics and microplastics in disposal sites.

Selected Publications

Yao Wang, C. Zhou, H. Zhang, S. Ma, Z. Lou, L. Raskin, S. Skerlos, F. Coulon, D. Zekkos, A. Hussain, A. Hussain, G. Lisak, M. Fang, K. Yin, H. He, Yuan Wang, and Xunchang Fei. Biogeochemical dynamics of major elements in municipal solid waste landfills can induce health risks for nearly 1 billion people. *One Earth* 8(9):101418, 2025.

Yao Wang, K. Yin, M. Fang, and Xunchang Fei. Underappreciated majority of solid waste: insufficient data and unclear impacts. *The Innovation* 6(3):100776, 2025.

Yao Wang, M. Fang, Y. Guo, X. Pi, Y. J. Wang, H. He, Z. Lou, K. Yin, and Xunchang Fei. Methane emissions differentially underestimated from landfills worldwide. *Nature Sustainability* 7:496-507, 2024.

Yao Wang, C. Zhou, Z. Lou, H. Zhang, A. Hussain, L. Zhan, K. Yin, M. Fang, and Xunchang Fei. Underestimated methane emission from solid waste disposal sites reveals missed greenhouse gas mitigation opportunities. *Engineering* 36:12-15, 2024.

Yao Wang, A. Ruiz-Acevedo, E. Rameez, V. Raghavan, A. Hussain, and Xunchang Fei. Toward sustainable waste management in small islands developing states: integrated waste-to-energy solutions in Maldives context. *Frontiers of Environmental Science & Engineering* 18(2):24-37, 2024.

Yao Wang, X. Lu, and Xunchang Fei. Property changes of conventional plastic waste mixed with municipal solid waste after 10-year degradation experiments simulating landfill conditions. *Journal of Hazardous Materials Letters* 2:100047, 2021.

Additional Publications

- K. Yin et al. Misbehaviour dominates GHG emissions from food loss and waste. *Nature Climate Change*, 2026.
- R. Ye et al. A novel semi-empirical model for multi-scenario estimations of nitrous oxide emissions from municipal solid waste landfills. *Environmental Science & Technology* 60(1):595-608, 2026.
- Z. Yang et al. Revealing hidden methane mitigation opportunities from legacy dumping and landfilling of municipal solid waste in rural China. *Journal of Environmental Management* 394:127244, 2025.
- V. Yadav et al. Gaps in quantifying environmental losses of plastics impede effective solutions. *Nature Reviews Materials*, 2025.
- Yao Wang** et al. Leaching model of an expanding coastal dumpsite considering climate change. *Waste Management* 200:114772, 2025.
- Z. Zhang et al. Impacts of regional socioeconomic statuses and global events on solid waste research reflected in six waste-focused journals. *Waste Management* 182:113-123, 2024.
- X. Pi et al. Global void ratio of municipal solid waste for compression indices estimation. *Waste Management* 160:69-79, 2023.
- X. Lu et al. Masses and size distributions of mechanically fragmented microplastics from LDPE and EPS under simulated landfill conditions. *Journal of Hazardous Materials* 445:130542, 2023.
- Xunchang Fei et al. The distribution, behavior, and release of macro- and micro-size plastic wastes in solid waste disposal sites. *Critical Reviews in Environmental Science and Technology* 53(3):366-389, 2023.
- Xunchang Fei et al. The long-term fates of land-disposed plastic waste. *Nature Reviews Earth & Environment* 3(11):733-735, 2022.
- Xunchang Fei, M. Fang, and **Yao Wang**. Climate change affects land-disposed waste. *Nature Climate Change* 11(12):1004-1005, 2021.

Grants and Projects

- Participant, "Breaking the ground" in Semakau Landfill: technological solutions for site investigation and material reuse, Singapore National Environment Agency.
- Participant, Long-term environmental behaviour for treated incineration bottom ash, Singapore National Environment Agency.
- Participant, Tackling microfibres at source, United Nations Development Program.

Teaching and Supervision

- Teaching Assistant, Nanyang Technological University: Solid Waste Management; Integrated Urban Solid Waste Management; Environmental Chemistry and Biology for Engineers; Water and Wastewater Engineering.
- Supervised five undergraduate research projects on landfill gases, dumpsite leachate, waste-to-energy, and food waste management.

Awards

- Chinese Government Award for Outstanding Self-Financed Students Abroad, 2025.
- Nanyang Technological University Research Scholarship, 2022-2026 and 2020-2022.
- NTU CEE Best Research Poster Award (Bronze), 2025.
- Tongji Scholarship of Excellence, 2016-2019.

Academic Service

- Young Editorial Board Member, The Innovation Geoscience and DeCarbon.
- Reviewer for Nature Communications, Journal of Environmental Management, and the 9th International Congress on Environmental Geotechnics.
- Invited talk at Frontiers of Environmental Science & Engineering Young Talent Forum, Tsinghua University, 2024.